

SAIVENKAT JILLA

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EDUCATION

University of Waterloo

Expected Apr. 2028

Bachelor of Computer Science, Artificial Intelligence Specialization, Minor in Combinatorics and Optimization

Average: 91%

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Capital One

Toronto, ON

- Drove credit-card acquisition via a **\$2M, 136M**-impression billboard campaign personalizing Quicksilver offers for drivers, using real-time weather/traffic/fuel signals + **computer vision** (fine-tuned 196-class **MobileNet**) to detect vehicle make
- Engineered an end-to-end interview-scheduling system (**Next.js, Go, PostgreSQL, AWS**) replacing **40+** spreadsheets across **2,590+** interviews and **7,770+** reach-outs/yr, saving **150 hrs/month** by automating slot matching and GCal integration
- Built a decision-support tool scoring prod-change risk **0–100** with blast-radius dependency graphs from **OpenAPI/IaC** files
- Shipped a **Go** CLI querying **Snowflake, Google Sheets**, and **CSVs** in natural language with per-connector access control

Machine Learning Engineer

Apr 2026 – Present

WAT.ai | SportsNext

Waterloo, ON

- Architected an end-to-end platform for a **JEPA**-style latent **world model** forecasting **multi-agent** soccer dynamics from broadcast video, spanning player/ball detection, a state builder, a **Mamba** transition backbone, and a **CEM/MPC** planner
- Developed a zero-data physics simulator and CI-validated **PyTorch** pipeline on a custom 5 Hz, 22-agent **JSON-Schema**

Undergraduate Research Assistant

Jan. 2026 – Apr. 2026

University of Waterloo | PI: Professor Srinivasan Keshav, Supervisor: Robin Young

Waterloo, ON

- Engineered **active learning** pipelines for GEDI biomass estimation, demonstrating that **spatial sampling** mitigates the cold-start problem by outperforming random baseline strategies by **27%** under extreme data starvation (**N=25**)
- Simulated **3,000+** smooth-to-chaotic 2D landscapes via **Gaussian Random Fields** to show which strategy wins where
- Architected a **Python/SciPy** evaluation suite to benchmark ML strategies, showing **Attentive Neural Processes (ANPs)** outperformed **Random Forests** by $> 3\times$ accuracy (R^2) in chaotic environments under data starvation ($< 1\%$ sample size)

Software Engineering Intern

Aug. 2025 – Jan. 2026

Marsh McLennan

Toronto, ON

- Automated case-handling by building an email classifier with **RAG-based** semantic search and a **RESTful Node.js** API
- Resolved **1,000+** exposed secrets by engineering a GitGuardian/JWT remediation pipeline to win an org-wide tournament
- Saved **100+ hr/wk** by automating **300+**-instance database deployments in GitHub Actions with rollback/secret handling
- Optimized reliability for **30+** apps by refining/documenting **multi-cloud** designs (OCI, AWS, Azure, GCP) with dev teams

Full-Stack Software Engineer

Jan. 2025 – Aug. 2025

The Mathematics Society of the University of Waterloo

Waterloo, ON

- Eliminated unreliable API dependencies for term lookups, stabilizing the interface via memoized utility functions
- Streamlined admin workflows by engineering a multi-select UI and RESTful backend endpoints for bulk exam management

Software Engineering Intern

Jan. 2025 – Apr. 2025

Global X Investments Canada

Toronto, ON

- Increased organic traffic to core financial pages by reducing load times by **15%** and refining technical **SEO**; achieved via API optimizations, frontend **lazy loading**, and metadata enhancements to maximize indexing and retention
- Drove a **32.4%** email open rate and scaled web presence by deploying **80+** pages/email blasts to support product launches
- Saved **100+ hr/yr** by automating **EN/FR** content synchronization and internal linking via custom **Python** and **PHP** scripts

PROJECTS

Spark: AI-Driven 3D Memory Capture | 1st Place, SummerHacks 2026 | Next.js, TypeScript, Python, Three.js, YOLO, Whisper

- Engineered a privacy-first system turning a real-world walk into an explorable 3D memory, pairing a camera rover with mobile **AR/LiDAR** sensing and fully local processing to index objects (**YOLO**), speech (**Whisper**), and routes for search
- Built a local-first **Gaussian splat** pipeline via phone pose, LiDAR depth, and GPS for reconstruction, rendered w/ **Three.js**
- Optimized object detection with multi-pass test-time augmentation and **Weighted Box Fusion** in place of NMS

CORnet-Mouse – biologically-constrained deep-RL model of mouse predator evasion | Unity ML-Agents, C#, Python, PyTorch

- Built a **Unity ML-Agents** 3D simulation where an agent forages while evading a looming predator amid distractors
- Tuned a **5-term reward function** (foraging, death, energy, false-alarm) to drive naturalistic freeze-vs-flee behavior
- Designed **CORnet-M**, a mouse-adapted **CORnet-S** variant with parallel, neuron-scaled visual pathways

TECHNICAL SKILLS

Languages: Python, Go, C/C++, C#, Java, TypeScript, JavaScript, HTML/CSS, SQL, Bash, R, MATLAB, PHP, VBA, Racket

Frameworks: pandas, NumPy, scikit-learn, Matplotlib, Scanpy, BeautifulSoup, React, Node, Express, Tailwind, Bootstrap

Tools/Infrastructure: Git, Docker, GitHub Actions, Linux, OCI, PostgreSQL/MySQL, MongoDB, Visual Studio, Figma, WordPress